



SAMP RQ4.2. Block 4A: Frozen Systematic Search Protocol, revision 3 (APPROVED, frozen 29 Sep 2026)



Status: APPROVED AND FROZEN, revision 3 (29 September 2026). Approved by Tagir Fabarisov. From this point every change goes through the deviation log (section 22). No Block-4 search has been run. Execution (4B) is done by a deterministic software pipeline built from this page and its query bank.

Open implementation question, passed to the pipeline builder: whether OpenAlex OQL accepts every construct in the OpenAlex forms, in particular truncation inside quoted phrases (for example `"dependence relation"`). It is checked against a recorded fixture before 4B; if a construct is rejected, the resolution is recorded in the deviation log before any retrieval, never applied silently.

Purpose. A search another researcher, or a program, can run from this page and its query bank alone. Block 3 was calibration; its candidate lists, IDs and statuses are not inputs. A method found in Block 3 becomes a Block-4 candidate only if this procedure finds it again.

Structure of Block 4. 4A this protocol · 4B retrieval · 4C deduplication, verification and screening. Detailed comparison is Block 5.

1. Scope

Main scope: RQ4.2 Problem 1, the representation of actor-side arrangements and their evolution: established relations; commitments, permissions, authority and power and other state types; participation and membership; dependencies between actors and arrangements; lifecycle and transformation; amendment, substitution, withdrawal, termination; timing and conditions; rules governing change; externally triggered change; organizational structure and governance relevant to arrangements.

Side registers, kept and routed, never discarded: P2 (collaborative design decisions, negotiation, consensus, voting, coordination, argumentation, architecture decision processes) and RQ4.3 (coupling of actor-side conditions to the technical architecture). A result may belong to several. Classification is not forced.

2. Search directions

Sixteen, from the Pass-3 calibration, IDs B4-D01 to B4-D16; F1-F14 are not used. The concepts covered are listed in revision 1 of this protocol and repeated in the query bank's "basis" column.

B4-D01 actor dependence, goals, assumptions, power and influence · D02 directed commitments and commitment lifecycle · D03 normative positions, permissions, obligations, exceptions and norm conflict · D04 institutional acts, constitutive rules, authority and institutional power · D05 roles, participation, membership and collective decision rights · D06 agreement and instrument force, bindingness, precision and consent · D07 relation identity, transfer, amendment, suspension, substitution and ending · D08 rule systems, meta-rules, rule levels and institutional rule change · D09 dynamic collaboration structure, membership change and organizational reconfiguration · D10 temporal and event semantics, deadlines, sequence, compliance and triggered effects · D11 staged processes, gates, milestones, reviews and staged commitment · D12 configuration baselines, engineering change control and approval routes · D13 options, delayed commitment, irreversibility, staged investment and lock-in · D14 external and non-agentive change and causal provenance · D15 technical system-of-systems structure, autonomy, ownership and reconfiguration · D16 informal governance, trust and relationship path dependence.

3. Three lenses per direction

Every direction has three pre-registered queries: Q-A semantic (the phenomenon, in Block-2 words), Q-B community (the vocabulary of the modeling communities and kinds of model), Q-C mechanism (how a field represents it). Two

cross-cutting queries (XC01 noncompliance and reparation; XC02 automatic vs discretionary) and four bridge queries (BR01–BR04, RQ4.3 register). Every direction gets the same treatment; no query was merged (the query bank explains the three cases where a merge could be argued). No known artifact, tool, project or method name appears in any query.

4. Query bank and syntax rules

The bank is on the subpage "Query Bank, revision 3" and in the machine-readable file `block4A_query_bank_rev3.yaml`. Query strings are shown in code formatting because Notion otherwise hides the asterisks of truncation wildcards.

Canonical form: Boolean with AND/OR, double-quoted phrases, `*` truncation, `organi?ation` for both spellings. Over-broad or wrong truncations were removed in revision 3: `norm*` (would match normal), `part*` (intended party or parties), `project*` (projection), `event*` (eventually), `rule*` (ruler), `power*`, and `"space program"` (would match programming) are written out as explicit OR lists.

Source	Form used	Adaptation rule
OpenAlex	OpenAlex Query Language: <code>title_and_abstract has (canonical)</code>	Boolean and <code>*</code> truncation are passed through as written; OpenAlex has no single-character wildcard, so every <code>organi?ation</code> form is written out as both spellings; run through the OQL/OQO endpoint, not the legacy search parameter
Scopus	<code>TITLE-ABS-KEY(canonical)</code>	Scopus supports <code>*</code> and <code>?</code> ; hyphenated phrases are searched as loose phrases in double quotes
Web of Science	<code>TS=(canonical)</code>	supports <code>*</code> and <code>?</code> ; same phrases
IEEE Xplore / ACM DL	canonical with <code>?</code> forms expanded to both spellings; wildcards inside phrases expanded to singular OR plural; bare <code>*</code> kept	IEEE supports <code>*</code> on bare terms (3 characters minimum before it) but not inside phrases and no <code>?</code>
HeinOnline	canonical	supports Boolean, <code>*</code> and <code>?</code> in advanced search; run in title-and-abstract fields
Google Scholar, SSRN	short form: at most eight terms, quoted phrases, OR, no wildcards	Scholar has no truncation and a length limit; the short form keeps the two most specific term groups of the canonical query
DBLP	short form with <code> </code> for OR	DBLP matches word prefixes by default and searches titles only

Closest-equivalent adaptation rule. Sources differ in syntax, fields, stemming and phrase handling, so no two forms of a query retrieve identical sets. The rule: keep the canonical Boolean structure and term groups; where a source lacks an operator, apply the adaptation in the table; where a source lacks truncation, write out the intended word forms; where a source limits length (Scholar), keep the two most specific term groups. Every adaptation is listed in the YAML per query, and every retrieval hit records the exact string sent, so differences between sources are auditable rather than assumed away.

5. Sources and allocation

Roles. OpenAlex: mandatory, open, reproducible, API-based broad scholarly source for all 54 queries. Scopus and Web of Science: independent curated bibliographic sources, mandatory. Domain sources (DBLP, IEEE Xplore, ACM DL, HeinOnline): mandatory by field as in the matrix. Google Scholar, SSRN and general web: supplementary; they are not run for the 54 queries as a matter of course, but invoked for a direction when a refinement calls for a recall source, as snowballing fallback, or for grey material; when invoked, the fixed depths of section 8 apply.

Access, confirmed 29 September 2026: University of Luxembourg interface and export access is confirmed for Scopus, Web of Science, IEEE Xplore, ACM DL and HeinOnline. Whether an API entitlement exists for any of them is an implementation check for the pipeline (API if entitled, otherwise the database's own export imported deterministically); it is not a blocker for freezing this protocol.

Direction	OpenAlex	Scopus	Web of Science	DBLP	IEEE / ACM	HeinOnline / SSRN	Google s
D01	M	M*	M*	M	S	–	S
D02	M	M*	M*	M	S (ACM)	–	S
D03	M	M*	M*	M	–	M*	S
D04	M	M*	M*	M	S (ACM)	S	S

Direction	OpenAlex	Scopus	Web of Science	DBLP	IEEE / ACM	HeinOnline / SSRN	Google S
D05	M	M*	M*	M	S	–	S
D06	M	M*	M*	S	–	M*	S
D07	M	M*	M*	S	–	M*	S
D08	M	M*	M*	M	–	M*	S
D09	M	M*	M*	M	M* (IEEE)	–	S
D10	M	M*	M*	M	S	–	S
D11	M	M*	M*	–	M* (IEEE)	–	S
D12	M	M*	M*	–	M* (IEEE)	–	S
D13	M	M*	M*	–	M* (IEEE)	–	S
D14	M	M*	M*	M	–	S	S
D15	M	M*	M*	S	M* (IEEE)	–	S
D16	M	M*	M*	M (Q-B only)	–	–	S
XC01, XC02	M	M*	M*	M	–	S	S
BR01-BR04	M	M*	M*	–	M* (IEEE)	–	S

M = mandatory; M* in the table also means mandatory (access confirmed; API or export is an implementation choice); S = supplementary, invoked as described under Roles; in the HeinOnline / SSRN column, HeinOnline is the mandatory legal source for D03, D06, D07 and D08 and SSRN is supplementary everywhere; the allocation applies to all three queries of the direction.

6. OpenAlex execution rules

- Query language: OpenAlex Query Language (OQL) with its query-object form (OQO), `title_and_abstract has ( ... )` carrying the canonical Boolean and `*` truncation directly; no manual synonym expansion; no year, language or type filter. The pipeline verifies the OQL grammar against a recorded fixture before 4B; any construct OpenAlex rejects is reported and resolved through the deviation log, never rewritten silently.
- Page size 100; cursor pagination in a fixed, supported ordering used only for reproducibility: ascending publication date, then OpenAlex work ID. Ordering has no role in inclusion, because every result of a query at or below the threshold is retrieved.
- Record for every run: query ID, exact query object sent, execution date and time (UTC), total count, number retrieved, pages, and the raw JSON of every page.
- Retrieve all results when the count is at most the threshold (section 8). If the count exceeds it, retrieval stops, the count is recorded, and the frozen refinement sequence of section 8 is applied by a person; the pipeline never invents a refinement.
- The API key is read from an environment variable or local secret store. It never appears in Notion, prompts, logs, configuration files or source code.

7. Document scope

- No lower year limit. Upper cutoff: the date the first 4B retrieval starts, written in the search log. The cutoff is enforced in normalization: any record with a publication date after the cutoff is kept in the raw data but flagged `after_cutoff` and excluded from S1.
- English-language material, or material with English title and abstract sufficient to screen; non-English full texts are recorded and flagged, not excluded at S1.
- Peer-reviewed papers, books and chapters, theses and formal technical reports can support a candidate. Standards only if they define a concrete representation. Project pages, repositories and documentation are supplementary evidence, never sufficient alone. Patents, practitioner pages and blogs are not candidate evidence except to identify an underlying scholarly artifact.

8. Result-processing rule

- **OpenAlex, Scopus, Web of Science, IEEE, ACM, DBLP, HeinOnline, SSRN:** screen all records when the query returns at most **300**. Confirmed: 300 keeps S1 exhaustive and, with dual automated screening (section 14), affordable (54 queries × up to 300 × up to 4 sources before deduplication). Above 300: never take the top N. Apply, in this order, recording the original count and every step: (1) restrict the free group `(model* OR formal* OR representation)` or its equivalent to the title field; (2) add the direction's Q-C mechanism group as an extra AND; (3) limit subject areas to computer science, engineering, decision sciences, social sciences, business, economics and law. If still above 300, split the query along its top-level OR groups and run each part. Every refined or split query gets a sub-ID (for example B4-Q07.r1).
- **Google Scholar:** when invoked, the first 100 results per query, relevance order, no year restriction; the first 100 "cited by" for forward-snowballing fallback.
- **General web:** when invoked, the first 20 results per query.
- Every count and depth inspected is logged.

9. Record model: hits and documents

- **Retrieval hit (B4-H00001 ...):** one per (query run, source, returned record): query ID, source, exact string sent, rank or page, date, source record ID (OpenAlex ID, Scopus EID, WoS accession, DOI, URL).
- **Unique document (B4-R00001 ...):** one per distinct work, created by deduplication before S1: DOI; OpenAlex ID; other source IDs; normalized title and year as fallback key; title; authors; year; venue; type; abstract; language.
- One document has many hits; every hit keeps its link to its document. Deduplication order: (1) DOI; (2) OpenAlex ID or other source ID; (3) normalized title (lower case, punctuation and stop words removed) plus year, with a manual check of near-duplicates.
- **Artifact (B4-A001 ...)** at S2; **family (B4-F01 ...)** at S3; **search aid (B4-S001 ...); expert seed (B4-X001 ...); registers P2-001 ... and B-001 ...**
- No Block-3 ID of any kind is used or cross-referenced in Block-4 data.

10. What is a candidate

A concrete modeling artifact that provides explicit constructs or semantics for representing at least one Problem 1 phenomenon: a modeling language, formalism or logic, metamodel, ontology, specification language, executable institutional or organizational framework, process, state or graph modeling method, or an architecture or enterprise modeling language in which arrangements are explicitly represented. A paper is evidence about an artifact. A theory, tool, standard or management framework is a candidate only when it defines an identifiable representation with constructs, relations or semantics that can be compared.

11. Families, versions and extensions

Each artifact, version, extension, tool implementation and domain specialization is recorded separately at S2 when its semantics are distinguishable, with lineage (base of, extension of, implementation of, specialization of). Merging into a family happens only at S3. A capability introduced by an extension is never attributed to the base without a base source. The evidence ledger keeps, per capability, the source and version that provides it.

12. Inclusion at S1 (recall-oriented)

A document proceeds if there is credible evidence that it defines a concrete modeling artifact; represents actors, arrangements, relations, norms, commitments, authority, membership, change, dependencies or another Block-4 direction; has identifiable constructs or semantics; could plausibly contribute to Problem 1. Not required: completeness, tooling, maturity, popularity, breadth.

13. Exclusion and routing codes

E1 survey or review only (to the search-aid ledger) · E2 descriptive theory or empirical framework only · E3 management practice without explicit representation · E4 tool without identifiable underlying model (attach as evidence if it implements a known artifact) · E5 duplicate publication or same-artifact evidence (link) · E6 no actor-side

arrangement content · E7 purely technical architecture artifact (route to RQ4.3 register) · E8 purely Problem 2 decision or coordination artifact (route to P2 register) · E9 insufficient information to verify (hold; one attempt to obtain the primary source) · E10 outside scope for another stated reason. A document may carry a routing code and still proceed in Problem 1.

14. Screening stages

S1, title and abstract, dual independent screening. Every unique document is screened by two independent screeners from two different language-model families, each with the frozen S1 question, codes and examples, and each blind to the other's decision and to all Block-3 material. The concrete model identifiers and prompt versions are recorded in the search log when screening starts and cannot change without a deviation-log entry. Rule: if either screener says proceed, the document proceeds. Disagreements are recorded with both rationales. Documents excluded by both screeners are audited by a person on a random sample of 10 percent, stratified by direction; if the audit finds more than 5 percent wrongly excluded in a stratum, that stratum is rescreened by a third screener. Documents marked "unsure" by either screener are held for human adjudication.

S2, artifact verification. Read enough of the primary source to state what the artifact is, its constructs, what it represents, version and family identity, and its register(s). Create B4-A. No concept or requirement scoring; that is Block 5.

S3, deduplication and family reconciliation. Resolve duplicate artifacts, aliases, versions, extensions and tool implementations (section 11). Create B4-F. Output of 4C: the candidate corpus.

15. Snowballing

Anchors: the primary defining source of every artifact that passed S2, and every search aid found by a Block-4 query. Block-3 items are not anchors unless Block 4 found them first. One backward round and one forward round (forward via OpenAlex [cited_by](#), Scopus or Web of Science; Scholar first 100 as fallback). Snowballed records get B4-H and B4-R IDs with provenance "snowball backward from B4-Axxx" or "snowball forward from B4-Sxxx" and are screened with the same criteria. No second round without a deviation-log entry.

16. Search-aid ledger

Reviews, surveys and comparison chapters are recorded as B4-S with field, scope, artifacts named, and whether used for backward or forward snowballing. They are not candidates unless they define an artifact.

17. Expert channel

The answers of Marco, Andreas and Maxime stay unopened until all pre-registered queries and both snowballing rounds are complete. Then each suggestion is verified, recorded as B4-X with provenance EXPERT SEED and "found by Block 4 before opening: yes / no"; every "no" is investigated and written down. Expert seeds may enter S2 through this separate channel; the database results are not rewritten.

18. Sentinel

None for Block 4. The Pass-3 sentinel already showed the calibrated terminology reaches the holistic system-of-systems and architecture design-space neighborhood. An independent holdout example, if Tagir provides one before 4B, is recorded here first.

19. RQ4.3 bridge and Problem 2

BR01-BR04 feed the RQ4.3 register; each hit gets one check: does the artifact also explicitly represent actor-side arrangements? If yes, it enters Problem 1 S1. Problem 2 has no queries of its own in Block 4; E8-routed documents are kept in the P2 register with title, source, artifact name if any, and reason.

20. Stopping rule

Every pre-registered query runs in every mandatory source. After all planned queries of a direction, refinement queries are allowed when terminology proved wrong, a new community appeared, or a mechanism is under-covered; each is

logged with its reason. Refinement stops after two consecutive refinement queries add no new artifact, no new representation mechanism and no new community or terminology. This is search stabilization, not exhaustiveness.

21. Block-3 independence and post-hoc comparison

During 4B and 4C: no Block-3 IDs, statuses or lists are visible to screeners or used to steer any query; no Block-3 provenance affects inclusion. After the 4C corpus is frozen, and only then, a post-hoc comparison is allowed and recorded on a separate page: which Block-4 artifacts also appeared in the Block-3 calibration; which calibration artifacts were not rediscovered, and why; which Block-4 artifacts are new relative to calibration. The comparison never alters the Block-4 result.

22. Protocol deviation log

Once frozen, any change is logged with: date; original rule or query; change; reason; before or after seeing relevant results; effect on earlier searches; whether earlier queries are rerun. No silent changes.

Deviation log. Frozen 29 September 2026. No deviations recorded.

Date	Original rule or query	Change	Reason	Before or after results	Effect on earlier searches	Rerun
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23. Authentication and access

The pipeline prefers, in order: official API access; legitimate exports from the database interface (CSV, RIS, BibTeX) imported by a deterministic importer; browser-assisted access only where the licence permits normal search and export, with the user logging in manually in a local browser profile. No passwords are stored; no MFA is bypassed; no access control or licence restriction is circumvented; no unattended scraping of licensed databases; session and cookie data stay local and outside version control. If a source cannot be used legitimately this way, it is run by hand and imported.

24. Order of execution

Directions D01 to D16, then XC, then BR; within a direction Q-A, Q-B, Q-C; within a query OpenAlex first, then Scopus, Web of Science, domain sources, Scholar, web. Fixed so that no direction is starved.

25. Templates

Search log (one row per query run): log ID · date-time UTC · query ID · direction · lens · source · exact string sent · original count · refinement steps and counts · depth inspected · hits created (B4-H range) · documents new after dedup · note.

Hit (B4-H): ID · document (B4-R) · query ID · source · exact string · rank or page · date · source record ID.

Document (B4-R): ID · DOI · OpenAlex ID · other IDs · normalized key · title · authors · year · venue · type · abstract · language · S1 decision A · S1 decision B · codes · adjudication · final S1.

Artifact (B4-A): ID · name as used by its authors · kind · defining source(s) · version · lineage · what it represents · constructs named · directions touched · register(s) · supporting documents · S2 date and person.

Family (B4-F): ID · members · relation · which member provides which capability, with source.

Search aid (B4-S), expert seed (B4-X), P2 register, RQ4.3 register, deviation log: as in revision 1, unchanged.

26. Decisions resolved in revision 3

- Access: interface and export access confirmed for Scopus, Web of Science, IEEE Xplore, ACM DL and HeinOnline; API entitlement is an implementation check.
- Thresholds: 300 structured, 100 Scholar and 20 web when invoked. Kept.
- Screening: two independent model families, 10 percent human audit of double-excludes, 5 percent rescreen trigger; model identifiers recorded when screening starts.

- Google Scholar, SSRN and web: supplementary, invoked on demand.
- HeinOnline mandatory for D03, D06, D07, D08; SSRN supplementary.
- OpenAlex ordering: ascending publication date, then work ID; page size 100; OQL/OQO.

27. Revision 3 changelog

1. OpenAlex queries use OQL/OQO `title_and_abstract has (...)` with the canonical Boolean and truncation, replacing the manual synonym expansion and the legacy search parameter.
2. OpenAlex page size 100.
3. OpenAlex ordering: publication date ascending, then work ID; ordering has no inclusion role.
4. Query fixes: `"space program"` written out so that it can never match programming (B4-Q32); `part*` replaced by `party OR parties` (B4-Q06); over-broad `norm*`, `project*`, `event*`, `rule*`, `power*` written out; final semantic audit of every source form recorded in the YAML.
5. "Identical lexical intention" replaced by the closest-equivalent adaptation rule (section 4).
6. Upper publication cutoff enforced in normalization (section 7).
7. Access marked confirmed for Scopus, Web of Science, IEEE Xplore, ACM DL, HeinOnline (section 5).
8. HeinOnline mandatory for D03, D06, D07, D08; SSRN supplementary.
9. Scholar and web are supplementary, not 54-query runs; depths apply when invoked.
10. Threshold 300 kept.
11. S1 frozen as two model families plus audit rules (section 14).
12. Coder denylist wording: specific candidate, project and tool names only; generic formalism and mechanism names permitted.

 SAMP RQ4.2. Block 4A: Query Bank, revision 2 (pre-registered, not run)

 SAMP RQ4.2. Block 4A: Query Bank, revision 3 (APPROVED, frozen 29 Sep 2026, not run)